

****Case study - Design and develop Nutrition Agentic AI Application using IBM Bob****

Prompt1 –

Design and develop Nutrition Agentic AI application with IBM Granite LLM . Agent should Collect user details: name, age, diet preference (veg/non-veg), medical history (e.g., diabetes, blood pressure), and location (city, country). Generate a personalized nutrition plan accordingly. Follow AI ethics, avoid harmful advice, and apply guardrails. Use a helpful, respectful tone (HAM). Include a clear medical disclaimer at the end.

Include agentic AI application using IBM Granite model to resolve users common queries. Design and develop web-based **Nutrition Agentic AI Application**.

Add Agent Assistance into .html file to accept user's query

Follow all phases of agentic AI SDLC to create the Nutrition Agentic AI application and save plan into .md file.

Create .bat file for user.

Task1 for user –

Create .env file with

IBM_API_KEY= "Paste_Your_IBMCloudAPIKEY"

IBM_PROJECT_ID="Paste_Your_IBMPROJECTID"

IBM_REGION= <https://us-south.ml.cloud.ibm.com>

PORT=3000

Prompt2 –

Use .env file and update all files .

Install necessary software and run the project from your end.

Task2 for user –

SignIn to your github account and Create new github repository for this project

Prompt3 –

Create readme.md file to deploy on github

Create git ignore file and requirement.txt file

Deploy all files into the github repository

<https://github.com/YogeshRaje/NutritionAgent-IBMBob.git>

Install necessary software and run the project from your end.

Prompt4 –

Create ppt which explains about project. Include the plan mentioned .md files data and all necessary output screenshots as per projects functionalities.

Prompt 5-

Deploy ppt file into the github repository

For eg - Deploy ppt file into the github repository

<https://github.com/YogeshRaje/NutritionAgent-IBMBob.git>

Prompt5 –

Create video with 150 seconds duration which explains about project. Include the plan mentioned .md files data and all necessary output screenshots as per projects functionality.

Here are practical options to get your 150 -seconds project video:

Option 1 — Export the PPT as Video (Fastest, 150 -seconds)

Task3 for user -

The presentation I just created (Nutrition_Presentation.pptx) can be exported as a video directly in PowerPoint:

Export as Video (3 clicks in PowerPoint)

1. In the open file: **File → Export → Create a Video**
2. Set quality: **Full HD 1080p**
3. Select: **"Use Recorded Timings and Narrations"**

4. Click **Create Video** → save as Nutrition_Video.mp4

Prompt6 –

Add voice over to .mp4 file

Prompt7 –

Deploy ppt file into the github repository

For eg- Deploy .pptx and .mp4 files into the github repository

<https://github.com/YogeshRaje/NutritionAgent-IBMBob.git>

Additional Task1 – Test the LLM

Prompt-

Test the Agent with all test parameters of LLM testing. Always show the test result of Agent correctness.

Additional Task2 – student can give prompt to change the theme and colour combination of Nutrition Application

Prompt-

Change the theme of website to white and light skyblue.

Note – Nutrition Agentic AI Application with IBM Bob is available on github

(Paste your github repository name)

For eg.

<https://github.com/YogeshRaje/NutritionAgent-IBMBob.git>

NutritionAgent – IBM Granite AI: Complete Delivery Report

 **Test Results — 76/76 PASSED (4m 38s)**

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NUTRITIONAGENT – IBM GRANITE AI | TEST CORRECTNESS REPORT

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Total : 76 | Passed : 76 | Failed : 0 | Skipped : 0

Status: ALL TESTS PASSED

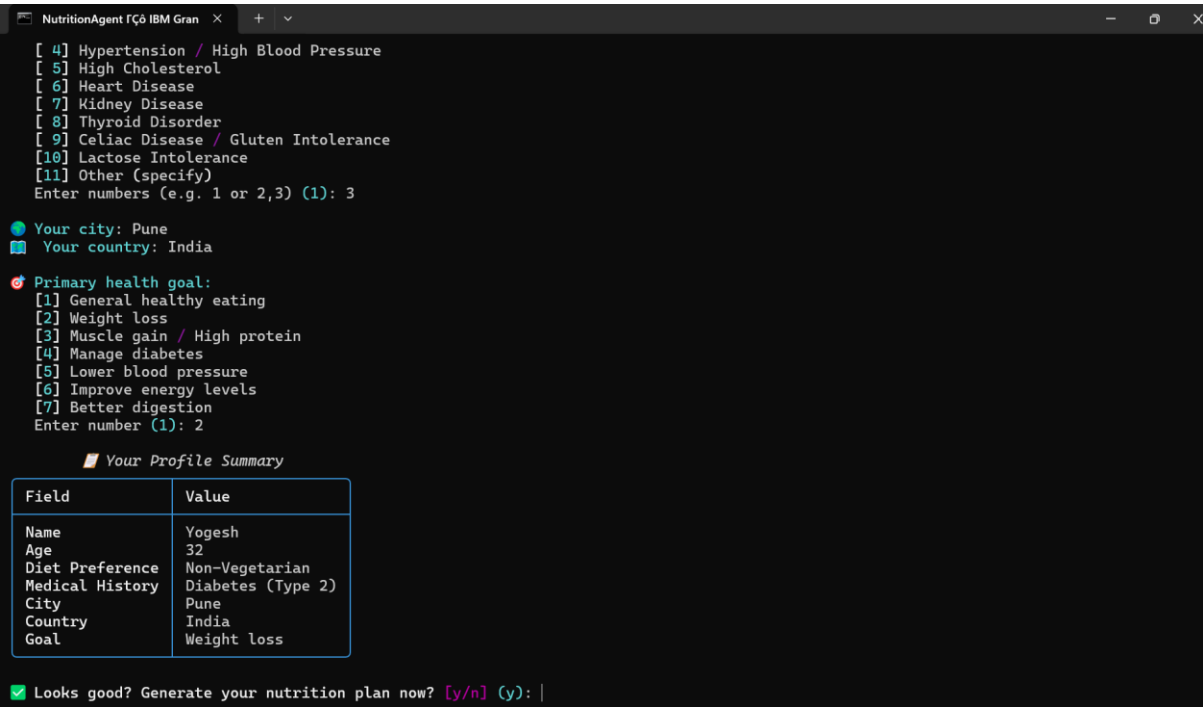
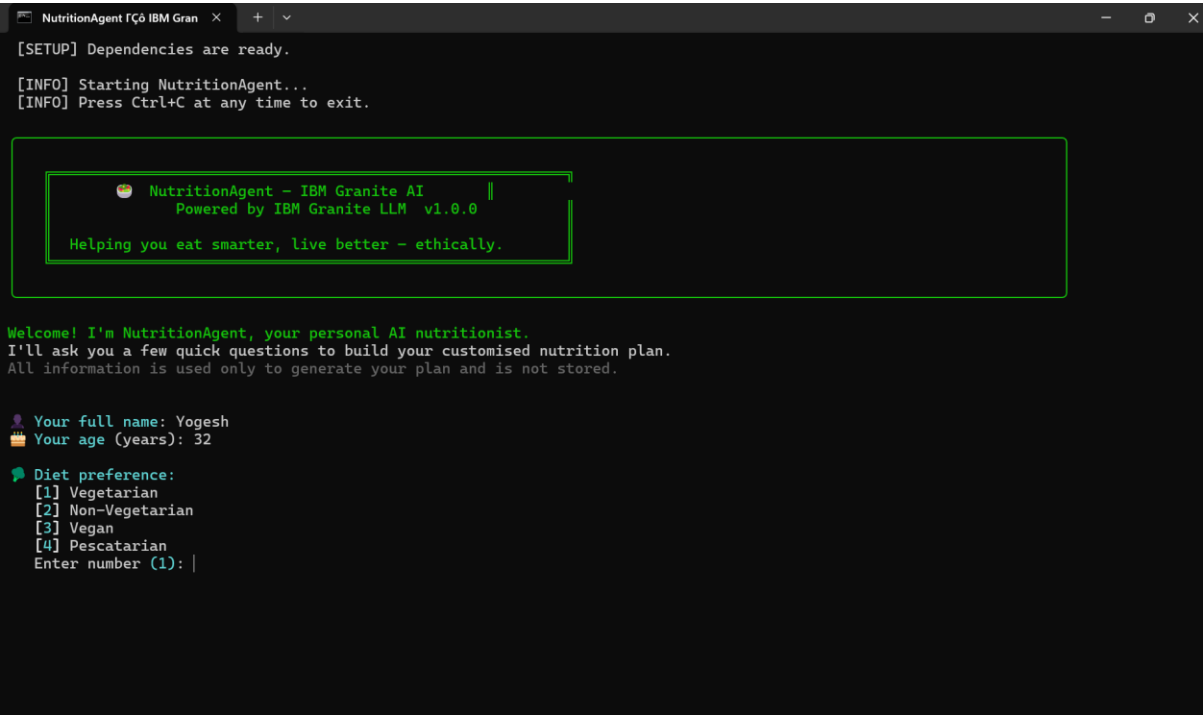
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Test Categories:

1. Configuration & Environment (10 tests)
2. Input Validation & Guardrails (16 tests – 11 harmful keyword params)
3. User Profile Schema (7 tests)
4. Filename & File Generation (6 tests)
5. Markdown Document Structure (7 tests)
6. LLM Integration (IBM Granite live) (10 tests)
7. HAM Compliance (4 tests)
8. Medical Safety (5 tests)
9. Output Quality (4 tests)
10. Edge Cases (7 tests)

Task- double click on run.bat file

Screenshot-



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# Personalised Nutrition Plan for Yogesh

## 1. Nutritional Overview

- Recommended Daily Calorie Range: ~1500-1700 kcal/day
  *This creates a modest caloric deficit (~500 kcal) aligned with safe weight loss while supporting blood sugar control.*

- Key Macronutrient Targets:
  - Carbohydrates: 45-55% of calories → ~200g/day (focus on low-GI sources)
  - Protein: 25-35% of calories → ~125-145g/day (supports muscle preservation and satiety)
  - Fat: 20-30% of calories → ~40-55g/day (emphasise heart-healthy fats)

- Micronutrients to Prioritise:
  - Chromium & Magnesium: Help improve insulin sensitivity.
  - Vitamin D & Calcium: Support bone health; many people with Type 2 diabetes have lower levels.
  - Fiber: Aids glycemic control and promotes fullness.

## 2. Weekly Meal Plan (Day-by-Day)

> General Guidance: Keep portion sizes moderate, use cooking methods like grilling, steaming, or light sautéing, and pair carbs with protein/fat to blunt glucose spikes.

### Monday
- Breakfast: 2 small egg bhurji (scrambled eggs with tomatoes, onions, coriander) + ½ whole-grain roti
- Mid-Morning Snack: Handful of roasted chana (chickpeas)
- Lunch: Grilled chicken tikka (~100 g) served with mixed veggie salad (cucumber, bell pepper, carrot) dressed with lemon-olive oil vinaigrette + ½ cup brown rice
- Afternoon Snack: Greek yogurt (plain, unsweetened, ½ cup) topped with few berries
- Dinner: Baked fish (hilsa or pomfret ~120 g) with palak paneer made with tofu instead of paneer, served alongside ½ cup millet (jowar/ragi) khichdi

### Tuesday
- Breakfast: Vegetable upma using semolina (sooji) cooked with lots of veggies (beans, carrots, peas) and a squeeze of lime

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